

FACULTY OF ENGINEERING AND TECHNOLOGY
University of Sindh, Jamshoro

BS (DS/SE) First Year Second Semester Examination 2023
DASC-320/SE23-318 Object Oriented Programming

(BS (DS) Morning) / (BS (SE) Evening) (Regular)

Date: 11-12-2023

Marks: 50

Time: 2 Hrs

→ Attempt carefully

- Instructions:
1. Attempt for only **50 Marks**
 2. Nobody is allowed to attempt more than **50 Marks**.
 3. Attempting more than **50** will result in Negative marking.
 4. Section I is compulsory, be precise while attempting.

SECTION I: Answer the following. (Attempt Any Five) → 10 Marks

- a) What do you know about multithreading? Is multithreading providing efficiency while java is losing its mobility because of multithreading. (Hint: Chapter 1)
- b) Can a super class be a super class of itself? Justify the statement and support your opinion.
- c) What are the recursive methods? Write a code of recursive method of your own.
- d) What is partial-implementation? Why is it needed?
- e) Can we utilize dynamic method dispatching and interface in performance-critical code? True or false, support your opinion with solid reasons.
- f) How does a program destroy an object that it creates? Discuss with reasons.
- g) Is there any similarity between the overridden methods in java and virtual functions in C++? What is the benefit of method overriding? What do you know about the variable overriding?
- h) "One interface multiple method", Identify and discuss aspect of polymorphism.
- i) Differentiate between abstract class and interface. Write simple code to demonstrate.
- j) "A non-static instance variable is illegal to access from inside of the static method". True or false; prove by writing a program.

SECTION II: Code a program

- Q.1 Explain in detail the benefits of instance variable hiding and variable overriding by writing programs. (10)
- Q.2 Java is considered the popular language for implementing the data structures algorithms; You are asked to write a complete code of stack in java by implementing the overflow and underflow situations. (10)
- Q.3 Write a program that uses command line arguments. (Assume: own input) (5)
- Q.4 • Write a program that satisfy the following situations.
- Two classes
 - 7 objects
 - 5 methods
 - 5 constructors
 - 2 methods in overloading
 - 2 constructors in overloading
 - 2 overridden constructs (10)
- Q.5 Why there is a need for Dynamic Method Dispatching? Elaborate the concept by writing a suitable example. (5)
- Q.6 • Can we define a class as protected? Yes, or No; Justify your answer by writing a simple program. (10)
- Q.7 • What will be the output of the program
- ```
class Dill {
 public static void main(String []args)
 {
 int a=09;
 System.out.println(a);
 }
}
```
- a=0909 ✓



**DEPARTMENT OF SOFTWARE ENGINEERING**  
**FACULTY OF ENGINEERING AND TECHNOLOGY**  
**UNIVERSITY OF SINDH, JAMSHORO**  
 Second Semester Examination 2023  
 BS (SWE) Part -I Regular (Evening)  
**Discrete Structures (SE-310)**

*Imprec*

Date: 13/12/2023

Marks: 50

Time 2:00 Hrs

Attempt any **FOUR** questions. All questions carry equal marks.

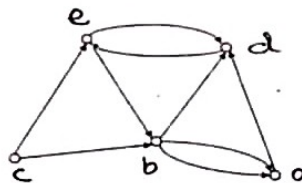
- Q1. (a) Discuss the principle of inclusion-exclusion in sets with example. 12  
 (b) Let A, B, and C be sets. Show that  

$$\overline{A \cup (B \cap C)} = (\overline{C \cup B}) \cap \overline{A}$$
 using set identities.  
 (c) Draw the Venn diagrams for each of these combinations of the sets A, B, and C  
 (i)  $(A \cap B) \cup (A \cap C)$  (ii)  $A \cap (B - C)$  (iii)  $A'$
- Q2. (a) Display the graph of the function  $f(x) = x^2$  from the set of integers to the set of integers.  
 (b) Let f be the function from  $\mathbf{R}$  to  $\mathbf{R}$  with  $f(x) = x^2$ . Is f invertible?  
 (c) Determine whether the function  $f(x) = x^2$  from the set of integers to the set of Integers is one-to-one.
- Q3. (a) Draw the graphs represented by the given adjacency matrices

$$\begin{pmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{pmatrix}$$

- (b) What does the degree of a vertex represent in the acquaintanceship graph, where vertices represent all the people in the world? What do isolated and pendant vertices in this graph represent? In one study it was estimated that the average degree of a vertex in this graph is 1000. What does this mean in terms of the model?
- Q4. (a) Determine the number of vertices and edges and find the in-degree and out-degree of each vertex for given directed graph.



- (b) Define bipartite graph. Draw the complete bipartite graphs  $K_{1,3}$ ,  $K_{4,4}$ .
- Q5. (a) Are the simple graphs with given adjacency matrices isomorphic?

(i)  $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$   $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

(ii)  $\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$   $\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

- Q6. Write short note on any **Three** of the following

- (1) Hand Shaking Theorem
- (2) Type of graphs
- (3) Type of functions
- (4) Applications of Discrete Structures
- (5) Cartesian Product and cardinality of sets







**FACULTY OF ENGINEERING AND TECHNOLOGY,  
UNIVERSITY OF SINDH JAMSHORO**

**2<sup>nd</sup> Semester Examination 2023      Date: 08-12-2023**

**Subject: SE23-320 Software Engineering**

**Class: BSSW P-I Evening Regular**

**Time: 2hours**

**Total Marks: 50**

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**Attempt any THREE (03) questions of the following, all questions carry equal marks**

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- Q1. (a) Define software. Draw the failure curve of software.  
(b) State and describe software generic process framework activities.
- Q2. (a) Distinguish between waterfall and spiral model.  
(b) Draw neat labelled diagram of incremental model. Describe working of incremental model with its advantages.
- Q3. (a) Distinguish between perspective process model and agile process model.  
(b) Describe extreme programming with proper diagram.
- Q4. (a) Explain six functions of requirement engineering process.  
(b) Explain what use cases and use case diagrams are, and how they can be used to capture the functional requirements for a software system.
- Q5. (a) Explain how software verification be achieved using unit testing and integration testing.  
(b) Describe the purpose of test plan for software validation process..

UNIVERSITY OF SINDH

INSTITUTE OF INFORMATION AND COMMUNICATION TECHNOLOGY

SUBJECT: English Composition and Comprehension

Date: 29/08/2023

CLASS: BS SWE PART I (EVENING)

Total Marks: 60

SEMESTER: FIRST 2023

Subject Teacher: **Ashfaq Mirani**

Time allowed: **2 Hours**

For Regular Students

Attempt and 4 questions. All questions carry equal marks

✓ Q1 write an Essay of Atleast 750 words on any one of the following topics .

A. Value of Artificial intelligence    B. Role of UN .    C. Amazon Market .

Q2. Write any two summaries on following lessons

A. Traffic Nightmare    B. The door swings open    C. Age of probot

Q3. Mention the key elements of CV and Resume writing with examples .

Q4. What is importance of Reading and write the types of reading .

Q5. What is sentence ? Describe its primary and secondary sentences with structure and examples

Q6. Define word classes in details and importance of Verb family with suitable examples.

and write the types of reading.

UNIVERSITY OF SINDH

INSTITUTE OF INFORMATION & COMMUNICATION TECHNOLOGY

SUBJECT: English Communication & Presentation Skills

CLASS: BS SWEE PART I (EVENING PROGRAME)

SEMESTER: Second 2023

**Paper: Midterm**

Subject Teacher: Ashfaq Mirani

Time allowed: **1 Hours**

Date: 19/10/2023

Total Marks: 20

**Attempt any two Question**

Q1: Discuss the difference between Composition and Comprehension skills and Communication and Presentation Skills?

Q2: How can one be a good presenter? Mention the Presentation protocols?

Q3: What is Punctuation? Write various kinds and use of punctuation?



1 examples

Q6. Define word classes in details and importance of Verb family with

Software Engineering  
PAKISTAN STUDIES (COMPULSORY) Course No: 302  
Part-I –Regular 1<sup>st</sup> SEMESTER 2023 (Evening Batch)

Time: 1 ½ Hour Attempt any TWO Questions. All Questions carry equal marks Max Marks 50

- Q.1 Highlight the circumstances that led to creation of Pakistan.
- Q.2 Discuss the election of 1937 and sufferings of Muslims under Congress Ministries.
- Q.3 Elaborate the initial problems faced by newly established state of Pakistan.
- Q.4 Write short notes on any TWO of following:
- a) Simla Deputation: 1906 Lord Minto
  - b) Lahore Resolution
  - c) Cabinet Mission Plan

**UNIVERSITY OF SINDH**  
**COMPARATIVE RELIGION & ISLAMIC CULTURE**  
**MIDTERM TEST 2<sup>ND</sup> SEMESTER 2023**

**CLASS: BS 1- IT- SOFTWARE/ DATA SCIENCE. MORNING AND EVENING**  
**SUBJECT: ISLAMIC STUDIES      TIME: 1 HOUR      MARX: 30**

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**ATTEMPT ANY TWO OF THE FOLLOWING QUESTIONS**

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- Q.1 Discuss in detail the existence of Allah.
- Q.2. Describe the significance of Tawheed in Islam.
- Q.3. What are the basic beliefs of Islam? Enlist them and discuss one of them in detail.

**UNIVERSITY OF SINDH, JAMSHORO (PAKISTAN)**

Department of Comparative Religion and Islamic Culture

Semester: Second

Final Test-2023 (**REGULAR**)

Class: **Software- Part-I (Morning & Evening)** Faculty Member: Dr. Sibghatullah Bhutto

Course: IS- 314

Subject: Islamic Studies (Compulsory)

Date: 15/12/2023

Time Allowed: 930-1130 AM

Max Marks: 50

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**NOTE:** Regular Students attempt any **THREE** Questions. Improvers and Those who could not appeared in Mid-Term Test attempt **FIVE** Questions. All Questions carry Equal Marks.

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- Q.1. Discuss in detail the arguments on the existence of God.
- ✓Q.2. Enlist the pillars of Islam and Explain the importance and benefits of Hajj in detail.
- Q.3. Elaborate the Attributes of a True believer in the light of Surah al-Muminoon.
- Q.4. What do you know about the concept of Knowledge and Ijtihad in Islam.
- Q.5. Elaborate the Fundamental Human Rights which Allah has given to all human beings.
- ✓Q.6. Discuss in detail the Rights of Women prescribed by Islam.
- ✓Q.7. Describe Jihad and explain its types in Islam.

*Best of Luck*



**UNIVERSITY OF SINDH**

**INSTITUTE OF INFORMATION & COMMUNICATION  
TECHNOLOGY**

**SUBJECT:** English Communication & Presentation Skills

**CLASS:** BSSW-1 (EVENING PROGRAMME)

**SEMESTER:** 2<sup>nd</sup> 2023

**Date: 06/12/2023**

**Total Marks: 60**

**Paper:** Final Term

**Subject Teacher:** Ashfaq Mirani

**Time allowed:** 2 Hours

**Regulars attempt any Four Questions & Improvers/Failures Five**

**Q1:** Write an Essay of 800 words on any one of the following.

- d) Impact of Prime Minister's Laptop Scheme of Students graduation level
- e) What university life teaches us
- f) Contribution of Co-curricular Activities in my academic career

**Q2:** Mention the Presentation steps on the topic of paragraph Writing & types.

**Q3:** Write Detailed Notes on any two of following

- d) Rules for Voice Change
- e) Formal & Informal Letter Writing
- f) Verb Formation

**Q4:** Write a narrative Paragraph on Cultural Day Celebration

**Q5:** Why Reading Comprehension is important? How to deal with it share your information

**DEPARTMENT OF SOFTWARE ENGINEERING  
FACULTY OF ENGINEERING AND TECHNOLOGY  
UNIVERSITY OF SINDH, JAMSHORO**

**First Semester (Evening) Regular Examination 2023**

**BS (SW) Part-I**

**CALCULUS AND ANALYTICAL GEOMETRY (SE23-306)**

Dated: 5/9/2023

Time Allowed: 2 Hours

Max Marks: 100

**NOTE: ATTEMPT ANY FIVE QUESTIONS**

| Q |                                                                                                                                                                                                                                                                                                                                                   | Marks |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | <p>Let <math>f(x) = \sqrt{x+1} + 4</math>.</p> <p>(i) Write the natural domain of <math>f</math>?</p> <p>(ii) <math>f(3) = ?</math></p> <p>(iii) <math>f(t^2 - 1) = ?</math></p> <p>(iv) <math>f(x) = 7</math> if <math>x = ?</math></p> <p>(v) The range of <math>f</math> is?</p>                                                               | 20    |
| 2 | <p>Let <math>f(x) = 3\sqrt{x} - 2</math> and <math>g(x) =  x </math>. In each part, give the formula for the function and state the corresponding domain.</p> <p>(i) <math>f + g</math> :</p> <p>(ii) <math>f - g</math> :</p> <p>(iii) <math>fg</math></p> <p>(iv) <math>f/g</math></p> <p>(v) <math>3f</math></p>                               | 20    |
| 3 | <p>Make sketches of the following graphs that show the period and amplitude.</p> <p>(a) <math>y = 3 \sin 2\pi x</math></p> <p>(b) <math>y = -3 \cos 0.5x</math></p> <p>(c) <math>y = 1 + \sin x</math></p>                                                                                                                                        | 20    |
| 4 | <p>Find a formula for <math>f^{-1}(x)</math>, and state the domain of the function <math>f^{-1}</math>.</p> <p>(i) <math>f(x) = 7x - 6</math></p> <p>(ii) <math>f(x) = 3x^3 - 5</math></p> <p>(iii) <math>f(x) = (x + 2)^4, x \geq 0</math></p>                                                                                                   | 20    |
| 5 | <p>Determine whether the following functions are continuous at <math>x = 2</math>.</p> <p><math>f(x) = \frac{x^2-4}{x-2},</math>      <math>g(x) = \begin{cases} \frac{x^2-4}{x-2}, &amp; x \neq 2 \\ 3 &amp; x = 2 \end{cases}</math>      <math>h(x) = \begin{cases} \frac{x^2-4}{x-2}, &amp; x \neq 2 \\ 4, &amp; x = 2 \end{cases}</math></p> | 20    |
| 6 | <p>A Find an equation for the tangent line to the parabola <math>y = x^2</math> at the point P (1, 1).</p> <p>B At what points, if any, does the graph of <math>y = x^3 - 3x + 4</math> have a horizontal tangent line?</p>                                                                                                                       | 10    |
| 7 | <p>A Use implicit differentiation to find <math>dy/dx</math> if <math>5y^2 + \sin y = x^2</math>.</p> <p>B Use implicit differentiation to find <math>d^2y/dx^2</math> if <math>4x^2 - 2y^2 = 9</math>.</p>                                                                                                                                       | 10    |
| 8 | <p>Evaluate each integral:</p> <p>(i) <math>\int \left[ 5x + \frac{2}{3x^5} \right] dx</math></p> <p>(ii) <math>\int \left[ x^{-\frac{1}{2}} - 3x^{\frac{7}{5}} + \frac{1}{9} \right] dx</math></p> <p>(iii) <math>\int \left[ \frac{10}{y^{\frac{3}{4}}} - \sqrt[3]{y} + \frac{4}{\sqrt{y}} \right] dy</math></p>                                | 20    |

# Faculty of Engineering and Technology

First Semester Final Examination of 2023

BS (Software Engineering) / Evening Part-I (Regular)

Course Title: Programming Fundamentals

Max. Marks: 60

Course Code: SENG-312

Time: 2 Hour

Note: Attempt any FIVE questions. All questions carry equal Marks

- Q1. (a) Briefly describe C++ Programming Language. And What do you mean by OOPs? Explain Structure of C++ Program? (b) Explain various data types used in C++. (c). Write down the difference between compiler and interpreter?
- Q2. (a) What is the difference between High Level and Low-Level language?  
(b) What is IF Statement and the terms ELSE and ELSE-IF with example?
- Q3. (a) Explain CIN and COUT statements in C++ with example?  
(b) How many types of operators? describe Logical and Arithmetic operators and create a Calculator program in which you use mathematical operations like addition, subtraction, multiplication and division.
- Q4. (a) How many types of Loops are in C++? Describe the Do While loop and create a program of Two Table using a while loop. (b) Write a program in C++ to display the number's digits in reverse order.
- Q5. (a) Write down a C++ program that covert Fahrenheit to Celsius scale. Formula  $(F-32)*5/9$ .  
(b) Write a program to print even numbers using a do-while loop.
- Q6. (a) The loop shown below has been written by an inexperienced C++ programmer. The behavior of the loop is not correctly represented by the formatting.
- ```
int n = 10;
while (n > 0)
n /= 2;
cout << n * n << endl;
```
- b. What is the output of the loop as it is written?
- Q7. What is a Database? What are tables and Fields? And What is the role of SQL?

"I am sure that you will do excellent in this exam. May ALLAH bless you with boundless success."

Faculty of Engineering and Technology

First Semester Final Examination of 2023

BS (Software Engineering) / Evening Part-I (Regular)

Course Title: Programming Fundamentals

Max. Marks: 60

Course Code: SENG-312

Time: 2 Hour

Note: Attempt any FIVE questions. All questions carry equal Marks

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Department of Software Engineering  
Faculty of Engineering and Technology  
UNIVERSITY OF SINDH

First Semester Examinations of 2023  
BS (Software Engineering) Part-I (Regular Evening)

**ELECTRONIC DEVICES AND CIRCUITS (SWEN-316)**

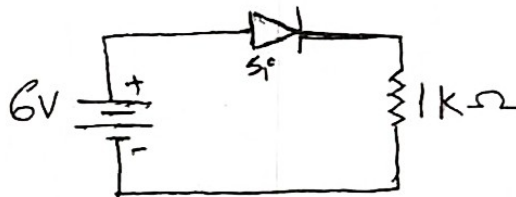
Monday, 28-08-2023

Time 2hrs

Max: Marks 50

Attempt any THREE questions. All questions carry equal marks

- Q. No: 1(a) Explain the characteristics <sup>of</sup> intrinsic and extrinsic semiconductor material. Why doping is essential for semiconductor devices?
- (b) Describe how P-type and N-type semiconductor materials are developed?
- Q. No: 2. What is rectification? Draw diagram of a half-wave rectifier circuit with input-out waveforms and describe the characteristics, operation and effect of turns ratio of primary and secondary windings.
- Q. No: 3: Differentiate any TWO of the following:
- (i) Barrier Potential and Depletion Layer
  - (ii) Valence band and Conduction band
  - (iii) Potentiometer and Rheostat
  - (iv) Characteristics of Close Circuit and Open Circuit
- Q. No: 4a. Explain the difference between Ideal diode and Complex diode with the help of equivalent circuit and characteristic curves.
- (b) Calculate forward current ( $I_F$ ) for and voltage drop across  $R_1$  for Ideal and practical diode for the circuit given below. Assume  $V_B = 0.65$  volts for practical diode.



- Q. No: 5(a) Describe the operation of Bipolar Junction Transistor (BJT) and draw appropriate symbols of BJT.
- (b) Draw BJT biasing circuits, operation regions, current and voltage parameters.
- Q. No: 6: Write short notes on any TWO of the following:
- (i) Breakdown region
  - (ii) Light Emitting Diode
  - (iii) Voltage Divider Circuit
  - (iv) Conductance and Resistance



FACULTY OF ENGINEERING AND TECHNOLOGY  
University of Sindh, Jamshoro

B.S (SE) Second Year, 1<sup>st</sup> Semester Examination 2023  
SENG - 314 Information and Communication Technology (ICT)  
(Evening) (Regular)

Date: 30-08-2023

Marks: 50

Time: 2 Hrs

→ Attempt Any Five Questions

- Q.1a) What do you know about the importance of ergonomics? How it is related to computers.
- Q.1b) ✓ How computers are related to society problems and how these computers can solve the numerous problems of the society. Explain in your own words.
- Q.2a) ✗ What are the components of computer? Explain in detail the main elements of computer.
- Q.2b) What are the categories of computers according to their size? Explain briefly.
- Q.3a) ✓ What is difference between primary memory and secondary storage?
- Q.3b) What are the categories of printers? Explain in detail.
- Q.4a) What is a processor and explain the terms microprocessor and nano processor.
- Q.4b) Explain in detail the various number systems you know.
- Q.5a) ✓ Explain the various types of Monitors.
- Q.5b) ✓ Explain the terms SSD, HDD, DDR, DIMM, SIMM along with their properties.
- Q.6a) ✓ What do you know about the input and output devices? List out and explain the various input and output devices used today.
- Q.6b) ✓ What is operating System? Explain its importance pertaining to software.
- Q.7 Write short notes on any **Three** of the following
- WWW and WWW2
  - Cloud computing
  - PDA
  - Blue Ray Disk
  - Utilities

IF U CAN DREAM IT !!!! U CAN MAKE IT HAPPEN